Logarithmic Functions & Properties

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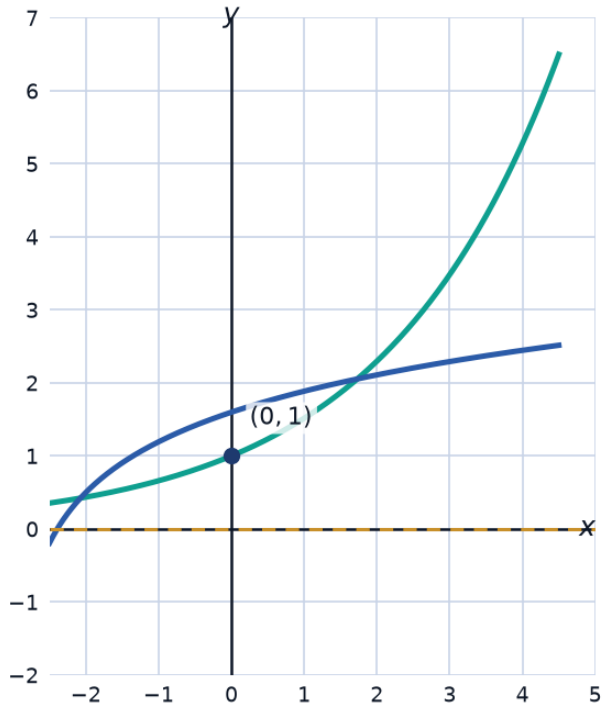
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Converting between exponential and logarithmic form

Rewrite $2^5 = 32$ in logarithmic form, and $\log_3 81 = 4$ in exponential form.

2 [Foundational] • Target: Converting between exponential and logarithmic form

Evaluate $\log_5 125$ and $\log_2 \frac{1}{8}$ without a calculator.



3 [Foundational] • Target: Properties of logarithms

Use logarithm properties to expand $\log\left(\frac{x^3\sqrt{y}}{z^2}\right)$ fully.

4 [Foundational] • Target: Properties of logarithms

Combine $2\log x - \frac{1}{3}\log y + \log z$ into a single logarithm.

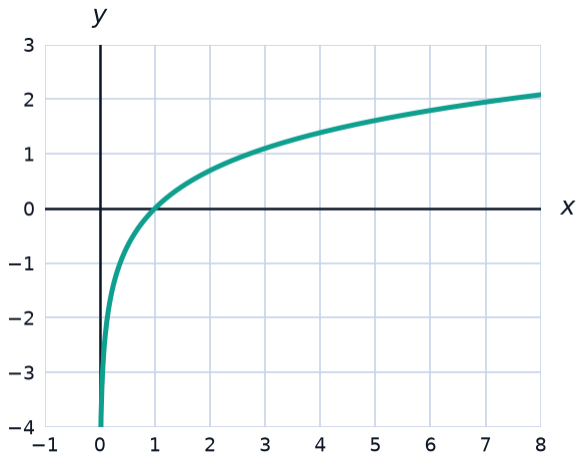
5 [Foundational] • Target: Properties of logarithms

Use the change of base formula to evaluate $\log_7 50$, to three decimal places.

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6 [Foundational] • Target: Visualizing logarithmic functions

The graph shows $f(x) = \ln(x)$, the natural logarithm.



a State the domain and the vertical asymptote shown in the graph.

b State the x-intercept shown.

7 [Foundational] • Target: Solving exponential equations using logarithms

Solve $5^x = 40$ for x, to three decimal places.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving exponential equations using logarithms

Solve $3(2)^{x-1} = 45$ for x, to three decimal places.

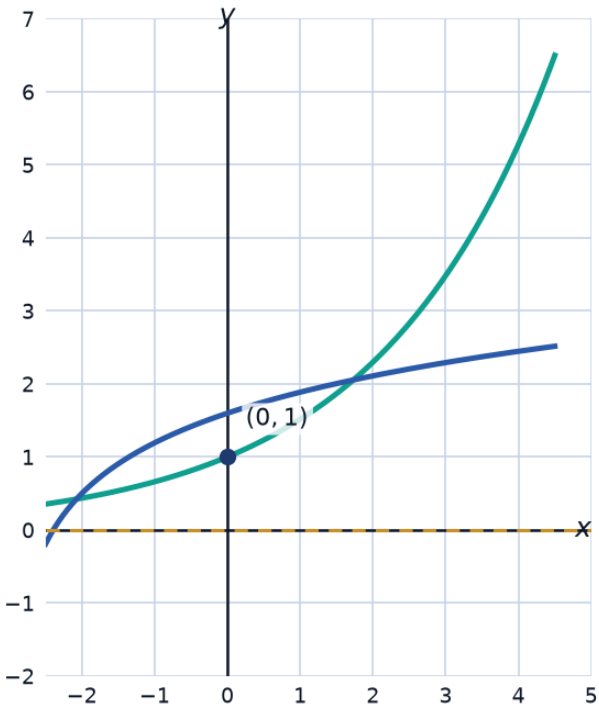
9 [Intermediate] • Target: Solving logarithmic equations

Solve $\log_2(x + 3) = 4$ for x, and check that the solution keeps the argument positive.

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10 [Intermediate] • Target: Solving logarithmic equations

Solve $\log(x) + \log(x - 3) = 1$ for x, checking both solutions against the domain.



11 [Intermediate] • Target: Applications: the pH scale and decibels

The pH of a solution is given by $\text{pH} = -\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration in mol/L. Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$ mol/L, to two decimal places.

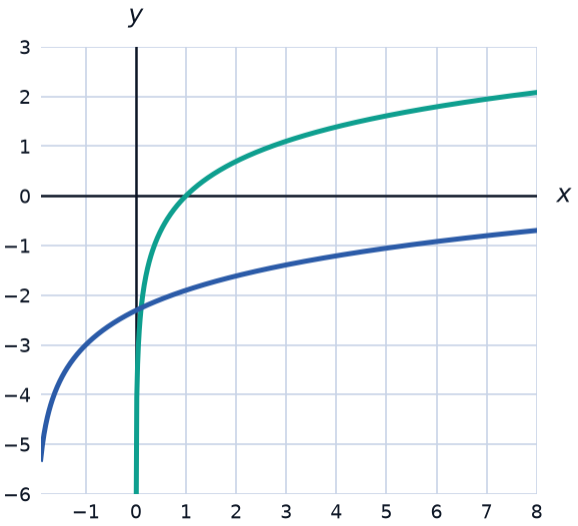
12 [Intermediate] • Target: Applications: the pH scale and decibels

Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. If one sound is 1000 times more intense than another, find the difference in their decibel levels.

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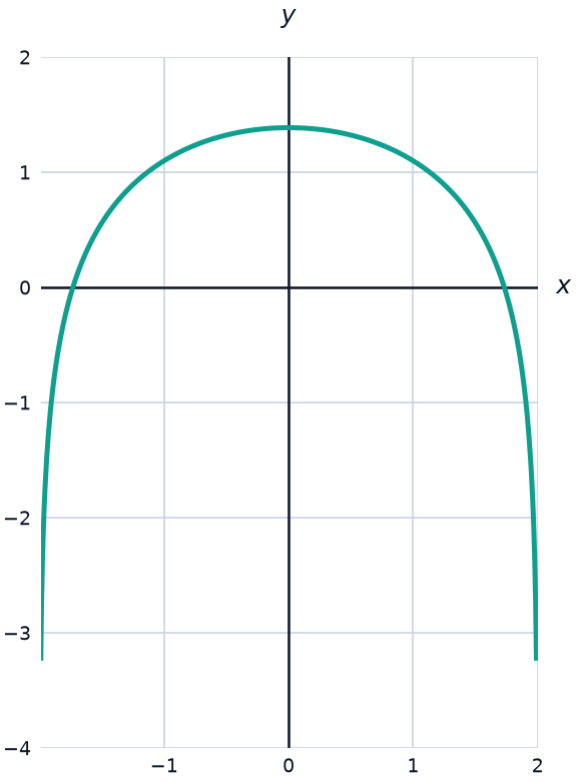
13 [Intermediate] • Target: Transformations of logarithmic functions

Describe how the graph of $g(x) = \ln(x + 2) - 3$ is transformed from the parent function $f(x) = \ln(x)$, and state its new vertical asymptote and domain.



14 [Intermediate] • Target: Transformations of logarithmic functions

Find the domain of $h(x) = \log(4 - x^2)$.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

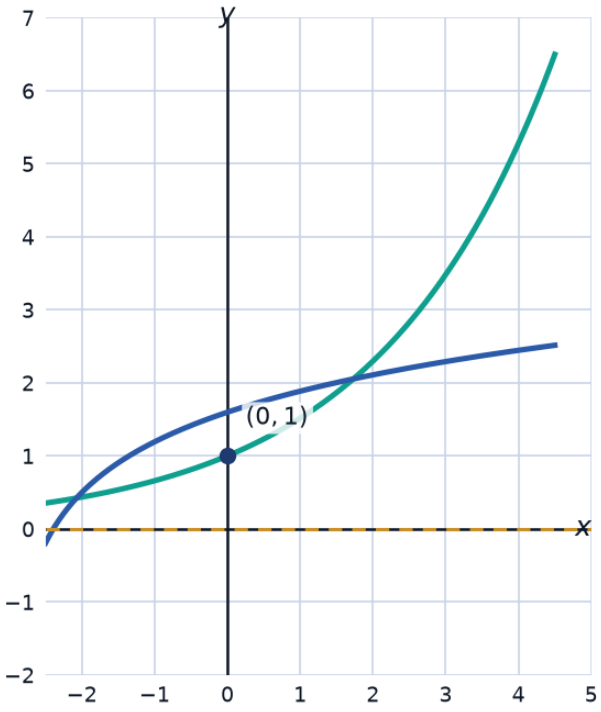
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15 [Challenging] • Target: Solving equations that combine exponentials and logarithms • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Solve $e^{2x} - 5e^{x} + 6 = 0$ by substituting $u = e^{x}$.

16 [Challenging] • Target: Solving equations that combine exponentials and logarithms • Cross-Lesson: 1.02 Rational Functions & Asymptotes

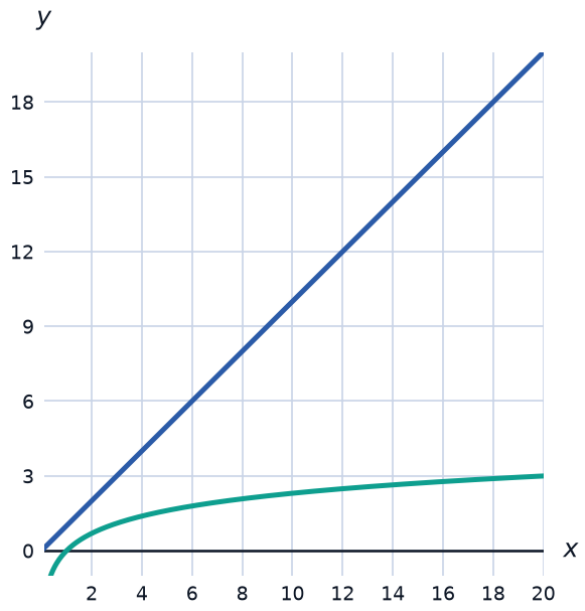
Solve $\log_2(x) + \log_2(x - 2) = 3$ for x, checking the domain.



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17 [Challenging] • Target: Comparing logarithmic growth to other function families • Cross-Lesson: 1.03 Rates of Change & Concavity

Explain why logarithmic functions grow much more slowly than linear functions for large x, and give a numerical comparison of $\log(x)$ and x at $x = 1000$.



18 [Challenging] • Target: Modeling with logarithms: the Richter scale • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Earthquake magnitude on the Richter scale is $M = \log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. An earthquake measures magnitude 7, while another measures magnitude 5. Find how many times more intense the magnitude-7 earthquake is.

19 [Challenging] • Target: Modeling with logarithms: the Richter scale • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

If an earthquake's intensity is $10{,}000$ times a reference intensity I_0, find its Richter magnitude.

20 [Challenging] • Target: Applications: the pH scale and decibels • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Solve the logarithmic equation $\log_4(x + 6) + \log_4(x - 6) = 3$, checking for extraneous roots.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Converting between exponential and logarithmic form

Evaluate $\log_2(2)$ without a calculator.

a. 1

b. -1

c. 0

d. 2

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22 [Foundational] • Target: Properties of logarithms

Evaluate $\log_6(36)$ without a calculator.

a. 2

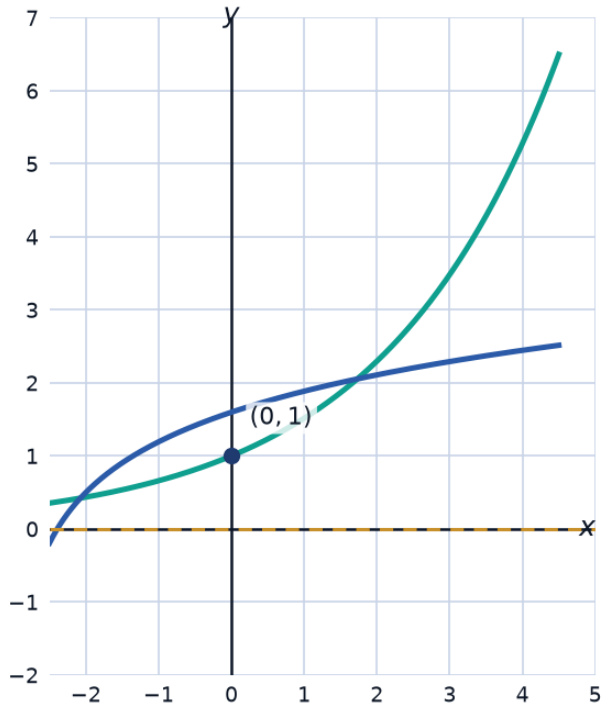
b. 1

c. -2

d. 3

23 [Foundational] • Target: Visualizing logarithmic functions

Evaluate $\log_4(16)$ without a calculator.



a. 3

b. 2

c. -2

d. 1

24 [Foundational] • Target: Solving exponential equations using logarithms

Evaluate $\log_4(64)$ without a calculator.

a. -3

b. 3

c. 4

d. 2

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25 [Foundational] • Target: Solving logarithmic equations

Evaluate $\log_4(4)$ without a calculator.

a. 2

b. 0

c. -1

d. 1

26 [Foundational] • Target: Applications: the pH scale and decibels

Evaluate $\log_2\left(\frac{1}{2}\right)$ without a calculator.

a. 0

b. 1

c. -2

d. -1

27 [Foundational] • Target: Transformations of logarithmic functions

Evaluate $\log_4\left(\frac{1}{256}\right)$ without a calculator.

a. -3

b. -5

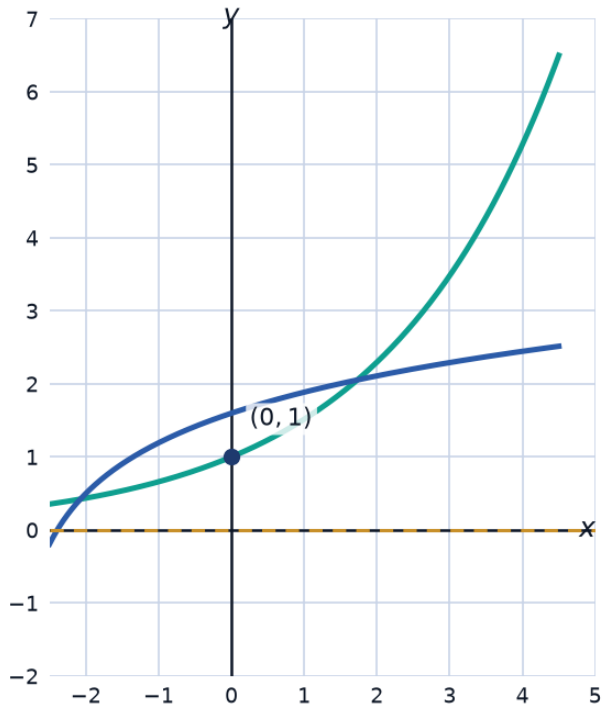
c. -4

d. 4

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28 [Foundational] • Target: Solving equations that combine exponentials and logarithms

Solve $\log_6(x) = 3$ for x.



a. 215

b. 217

c. 216

d. 218

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Comparing logarithmic growth to other function families

Solve $\log_5(x - 5) = 3$ for x.

a. 129

b. 130

c. 120

d. 131

30 [Intermediate] • Target: Modeling with logarithms: the Richter scale

Solve $\log_4(x + 9) = 1$ for x.

a. 13

b. -4

c. -5

d. -6

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31 [Intermediate] • Target: Definition of Logarithms and Inverses

Solve $\log_3(x+4)=2$ for x.

a. 13

b. 4

c. 5

d. 6

32 [Intermediate] • Target: Logarithmic Properties & Condensing Expressions

Solve $\log_6(x-3)=1$ for x.

a. 8

b. 9

c. 10

d. 3

33 [Intermediate] • Target: Solving Exponential and Logarithmic Equations

Solve $\log_2(x+7)=1$ for x.

a. -5

b. 9

c. -4

d. -6

34 [Intermediate] • Target: Logarithmic Scales and Applications

Solve $\log_6(x-6)=2$ for x.

a. 42

b. 43

c. 30

d. 41

35 [Intermediate] • Target: Converting between exponential and logarithmic form

Solve $9^x=165$ for x, to three decimal places.

a. 2.824

b. 2.424

c. 2.224

d. 2.324

36 [Intermediate] • Target: Properties of logarithms

Solve $6^x=90$ for x, to three decimal places.

a. 2.411

b. 2.511

c. 2.611

d. 3.011

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37 [Intermediate] • Target: Visualizing logarithmic functions

Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^6 times more intense than another, find the difference in their decibel levels.

a. 50

b. 70

c. 60

d. 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving exponential equations using logarithms • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^2 times more intense than another, find the difference in their decibel levels.

a. 20

b. 10

c. 2

d. 30

39 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Solve $\log_{21}(x) + \log_{21}(x-4) = 1$ for x, checking both solutions against the domain.

a. 8

b. -3

c. 7

d. 4

40 [Challenging] • Target: Applications: the pH scale and decibels • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Solve $\log_6(x) + \log_6(x-5) = 1$ for x, checking both solutions against the domain.

a. 6

b. 7

c. -1

d. 5

41 [Challenging] • Target: Definition of Logarithms and Inverses • Cross-Lesson: 4.01 Inverse & Composite Functions

Evaluate $\log_4\left(\frac{1}{64}\right)$.

a. -3

b. 3

c. $-\frac{1}{3}$

d. 16

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42 [Challenging] • Target: Definition of Logarithms and Inverses • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Find the inverse function of $g(x) = 3 + 2\ln(x - 1)$.

a. $g^{-1}(x) = e^{(x-3)/2} + 1$

b. $g^{-1}(x) = e^{(x-3)/2} - 1$

c. $g^{-1}(x) = 2e^{x-3} + 1$

d. $g^{-1}(x) = e^{2(x-3)} + 1$

43 [Challenging] • Target: Logarithmic Properties & Condensing Expressions • Cross-Lesson: 4.03 Parametric Equations & Modeling

Using the change of base formula, express $\log_5 12$ in terms of natural logarithms.

a. $\frac{\ln 12}{\ln 5}$

b. $\frac{\ln 5}{\ln 12}$

c. $\ln 12 - \ln 5$

d. $\ln\left(\frac{12}{5}\right)$

44 [Challenging] • Target: Solving Exponential and Logarithmic Equations • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Solve for x: $\log_2(x) + \log_2(x - 2) = 3$.

a. 4

b. 4 and -2

c. 2

d. 8

45 [Challenging] • Target: Logarithmic Scales and Applications • Cross-Lesson: 1.02 Rational Functions & Asymptotes

The pH of a chemical solution is given by $\text{pH} = -\log[\text{H}^+]$. If the hydrogen ion concentration is 1.0×10^{-5} M, what is the pH?

a. 5.0

b. -5.0

c. 1.0

d. 9.0

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